

A Technical Introduction to LLaMA language model

Section 1

Software developer Georgi Gerganov released llama.cpp as open-source on March 10, 2023. It's a re-implementation of Llama in C++, allowing systems without a powerful GPU to run the model locally. The llama.cpp project introduced the GGUF file format, a binary format that stores both tensors and metadata. The format focuses on supporting different quantization types, which can reduce memory usage, and increase speed at the expense of lower model precision. Llamafile created by Justine Tunney is an open-source tool that bundles llama.cpp with the model into a single executable file. Tunney et al. introduced new optimized matrix multiplication kernels for x86 and ARM CPUs, improving prompt evaluation performance for FP16 and 8-bit quantized data types.

Section 2

Reception Wired describes the 8B parameter version of Llama 3 as being "surprisingly capable" given its size. The response to Meta's integration of Llama into Facebook was mixed, with some users confused after Meta AI told a parental group that it had a child. The release of Llama models has sparked significant debates on the benefits and misuse risks of open-weight models. Such models can be fine-tuned to remove safeguards, notably by cyber criminals, until they comply with harmful requests. Some experts contend that future models may facilitate causing damage more than defending against it, for example by making it relatively easy to engineer advanced bioweapons without specialized knowledge. Conversely, open-weight models can be useful for a wide variety of purposes, including for safety research. Open Source Initiative head Stefano Maffulli criticized Meta for describing Llama as open-source, saying that it was causing confusion among users and "polluting" the term.

Section 3

Initial release The first version of Llama (stylized as LLaMA and sometimes referred to as Llama 1) was announced on February 24, 2023, via a blog post and a paper describing the model's training, architecture, and performance. The inference code used to run the model was publicly released under the open-source GPLv3 license. Access to the model's weights was managed by an application process, with access to be granted "on a case-by-case basis to academic researchers; those affiliated with organizations in government, civil society, and academia; and industry research laboratories around the world". Llama was trained on only publicly available information, and was trained at various model sizes, with the intention to make it more

accessible to different hardware. The model was exclusively a foundation model, although the paper contained examples of instruction fine-tuned versions of the model. Meta AI reported the 13B parameter model performance on most NLP benchmarks exceeded that of the much larger GPT-3 (with 175B parameters), and the largest 65B model was competitive with state of the art models such as PaLM and Chinchilla.

Section 4

Scout: 17 billion active parameter model with 16 experts, context window of 10M, with 109B parameters in total. **Maverick:** 17 billion active parameter model with 128 experts, context window of 1M, with 400B parameters in total. The Behemoth model was also announced, but was not released. Meta claimed it was a 288 billion active parameter model with 16 experts and around 2T parameters in total; it was still in training when Scout and Maverick were released. Maverick was codistilled from Behemoth, while Scout was trained from scratch. The training data included publicly available data, licensed data, and Meta-proprietary data such as publicly shared posts from Instagram and Facebook and people's interactions with Meta AI. The knowledge cutoff was August 2024. Meta claimed in its release announcement that Llama 4 bested GPT-4o's score on the LMArena AI benchmark. The company also stated that Llama 4's benchmark score was achieved using an unreleased "experimental chat version" of the model that was "optimized for conversationality", which differed from the version of Llama 4 released to the public. LMArena indicated that it would change its policies to prevent this incident from reoccurring, and responded, "Meta's interpretation of our policy did not match what we expect from model providers. Meta should have made it clearer that 'Llama-4-Maverick-03-26-Experimental' was a customized model to optimize for human preference." Some users criticized Meta on social media for its use of a separate model version tailored for benchmarking, and some additionally accused Meta of training Llama 4 on test sets to further boost its benchmark scores, which Meta denied.

Section 5

Space Booz Allen Hamilton deployed Meta's Llama 3.2 model aboard the International Space Station (ISS) National Labs as part of a project called Space Llama. The system runs on Hewlett Packard Enterprise's Spaceborne Computer² and leverages Booz Allen's A2E2 (AI for Edge Environments) platform, using NVIDIA CUDA² accelerated computing. Space Llama demonstrates how large language models can operate in disconnected, constrained environments such as space, enabling astronauts to retrieve and summarize documents using natural-language queries, even without internet connectivity.

Section 6

Llama ("Large Language Model Meta AI" serving as a backronym) is a family of large language models (LLMs) released by Meta AI starting in February 2023. Llama models come in different sizes, ranging from 1 billion to 2 trillion parameters. Initially only a foundation model, starting with Llama 2, Meta AI released instruction fine-tuned versions alongside foundation models. Model weights for the first version of Llama were only available to researchers on a case-by-case basis, under a non-commercial license. Unauthorized copies of the first model were shared via BitTorrent. Subsequent versions of Llama were made accessible outside academia and released under licenses that permitted some commercial use. Alongside the release of Llama 3, Meta rolled out Meta AI, an AI assistant built on Llama. Meta AI has a dedicated website and is available on Facebook and WhatsApp. The latest version is Llama 4, released in April 2025. In April 2026, Meta Superintelligence Labs released Muse Spark as a replacement for Llama.

Section 7

Leak On March 3, 2023, a torrent containing Llama's weights was uploaded, with a link to the torrent shared on the 4chan imageboard and subsequently spread through online AI communities. That same day, a pull request on the main Llama repository was opened, requesting to add the magnet link to the official documentation. On March 4, a pull request was opened to add links to HuggingFace repositories containing the model. On March 6, Meta filed takedown requests to remove the HuggingFace repositories linked in the pull request, characterizing it as "unauthorized distribution" of the model. HuggingFace complied with the requests. On March 20, Meta filed a DMCA takedown request for copyright infringement against a repository containing a script that downloaded Llama from a mirror, and GitHub complied the next day. Reactions to the leak varied. Some speculated that the model would be used for malicious purposes, such as more sophisticated spam. Some have celebrated the model's accessibility, as well as the fact that smaller versions of the model can be run relatively cheaply, suggesting that this will promote the flourishing of additional research developments. Multiple commentators, such as Simon Willison, compared Llama to Stable Diffusion, a text-to-image model which, unlike comparably sophisticated models which preceded it, was openly distributed, leading to a rapid proliferation of associated tools, techniques, and software.